ANPR SOFTWARE PROJECT

Time & Cost Estimation

We have reviewed the provided ANPR MVP requirement. The proposed solution will run on-premises and will cover vehicle recognition, number-plate processing, vehicle records, whitelist/blacklist rules, events and a basic web dashboard.

Proposed Approach

Instead of using dedicated ANPR cameras, we will use normal IP surveillance cameras with RTSP video streams. The video will be processed by the application itself.

Surveillance Camera → RTSP Stream → Vehicle Detection → Number Plate Detection → OCR → Rules → Event Storage → Dashboard

AI / Computer Vision Work

Since we are **not using dedicated ANPR cameras**, the camera will only provide the video stream. The vehicle and number-plate recognition will need to be handled on the software side.

This means additional AI/computer-vision work is required for vehicle detection and number-plate detection, followed by OCR to read the detected plate.

The application will process the surveillance-camera frames, detect vehicles, identify the number-plate area, process the plate image and extract the plate number. The extracted information will then be passed to the rules and event-processing part of the application.

Technology

Part	Technology
Camera input	IP surveillance camera / RTSP
Video processing	OpenCV / FFmpeg
Vehicle detection	YOLO-based computer vision
Number-plate detection	YOLO-based computer vision
Number-plate reading	OCR
Backend	Python + FastAPI
Database	PostgreSQL
Dashboard	React
Live updates	WebSocket
Image storage	Local storage with database reference

Work Included

• RTSP surveillance camera connection and video processing.

• AI-based vehicle detection from the surveillance-camera stream.

• AI-based number-plate detection from vehicle/frame images.

• OCR-based number-plate reading.

• Vehicle records and event storage.

• Whitelist and blacklist checking.

• Basic event and alert handling.

• Basic web dashboard with latest events and vehicle information.

• Basic search and user access.

• Local PostgreSQL database and image storage.

• Local/on-premises deployment.

Time & Cost Estimation

The estimated development time for the main implementation is **4 weeks**. An additional **Week 5** is kept for minimal required changes and edge setup assistance.

Week	Work	Cost
Week 1	Project setup, database, RTSP connection and initial AI vehicle/plate detection work.	Rs 7,500
Week 2	OCR, plate processing, event creation, vehicle records and whitelist/blacklist rules.	Rs 7,500
Week 3	Dashboard, live/latest events, search and user access.	Rs 7,500
Week 4	Testing, AI/camera tuning, bug fixing and local deployment.	Rs 7,500
Week 5	Minimal required changes and edge setup assistance.	—
Total		**Rs 30,000**

Week 5 is an additional support week for minimal required changes and edge setup assistance.

Main Development Cost: Rs 30,000

Requirement from Client

A working IP surveillance camera with RTSP access and a suitable view of the vehicles/number plates will be required for development and testing. Since normal surveillance cameras are being used, video quality, camera position, lighting and distance from the vehicle will affect number-plate reading.

The server/PC, camera, network equipment and other physical hardware are not included in the **Rs 30,000** software development cost.

Review & Confirmation

Please review the above approach, timeline and cost estimate and let us know if any changes or revisions are required. The quoted **Rs 30,000** is negotiable based on the final agreed requirements and scope.

Main Development: 4 weeks   Cost: Rs 30,000 (Negotiable)

Thank you.